



Swiggy Business Performance Analysis

Transforming Raw Data into Actionable Insights

TOOLS USED

SQL Server • Python • Excel • Power BI

KEY FOCUS AREAS



Sales Analysis



Restaurant Performance



Location Insights



Customer Ordering
Trends



Interactive Reporting

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Internal All
Employees

Project Overview

An end-to-end analytics pipeline, from raw order data to a decision-ready dashboard.

Objective

Build a complete data analytics pipeline that transforms raw Swiggy order data into meaningful business insights — cleaning and validating the data, modeling it into a star schema, and delivering interactive dashboards that help stakeholders understand sales, restaurant, and location performance.

Dataset Information

- 5 linked source tables: fact_orders, dim_restaurant, dim_location, dim_dish, dim_date
- 197K+ orders merged into a single analytical dataset
- Coverage across 25+ Indian states and union territories

Business Questions Addressed

- Which days and months drive the most revenue?
- Which restaurants and states generate the most sales?
- How does demand differ between veg and non-veg orders?
- How consistent is customer satisfaction (ratings) over time?
- What is the average order value, and how does it vary?
- How does quarterly performance trend across the year?

SQL Analysis

Data cleaning & validation layer — Query screenshots below.

```
70  ### Data Cleaning Queries
71  ###Check Null Values
72  • select * from fact_orders where order_id IS NULL or date_id
73  IS NULL or location_id IS NULL
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

order_id	date_id	location_id	restaurant_id	food_id	price	rating	rating_count
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Data Cleaning

```
76  • select * from dim_restaurant where restaurant_id IS NULL;
77  • select * from dim_location where location_id IS NULL;
78  • select * from dim_dish where dish_id IS NULL;
79  • SELECT *
80  FROM dim_dish
81  WHERE dish_id IS NULL
82  OR category IS NULL
83  OR dish_name IS NULL;
84
85  ### COUNT THE VALUES
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

dish_id	category	dish_name
NULL	NULL	NULL

NULL Checks

```
102  #####To find duplicates
103  • SELECT dish_id, category, dish_name, COUNT(*) AS cnt
104  FROM dim_dish
105  GROUP BY dish_id, category, dish_name
106  HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

dish_id	category	dish_name	cnt
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Duplicate Removal

```
136  • select * from fact_orders f join dim_date d on f.date_id=d.date_id
137  join dim_dish di on f.food_id=di.dish_id join dim_location l on l.location_id=f.location_id
138  join dim_restaurant r on f.restaurant_id=r.restaurant_id;
139
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows: |

order_id	date_id	location_id	restaurant_id	food_id	price	rating	rating_count	date_id	order_date	Year	Month	Month_name	Quarter	w
1	1	1	1	1	133.90	4.0	0	1	2029-06-25	2029	6	June	2	25
1	1	1	1	1	133.90	4.0	0	1	2029-06-25	2029	6	June	2	25
1	1	1	1	1	133.90	4.0	0	1	2029-06-25	2029	6	June	2	25
1	1	1	1	1	133.90	4.0	0	1	2029-06-25	2029	6	June	2	25
1	1	1	1	1	133.90	4.0	0	1	2029-06-25	2029	6	June	2	25

Result 23 x

Joins

```
144  ## Total_sales
145  • SELECT CONCAT(
146  FORMAT(SUM(price)/1000000, 2),
147  ' INR Millions'
148  ) AS total_sales
```

Result Grid | Filter Rows: | Export: | Wr

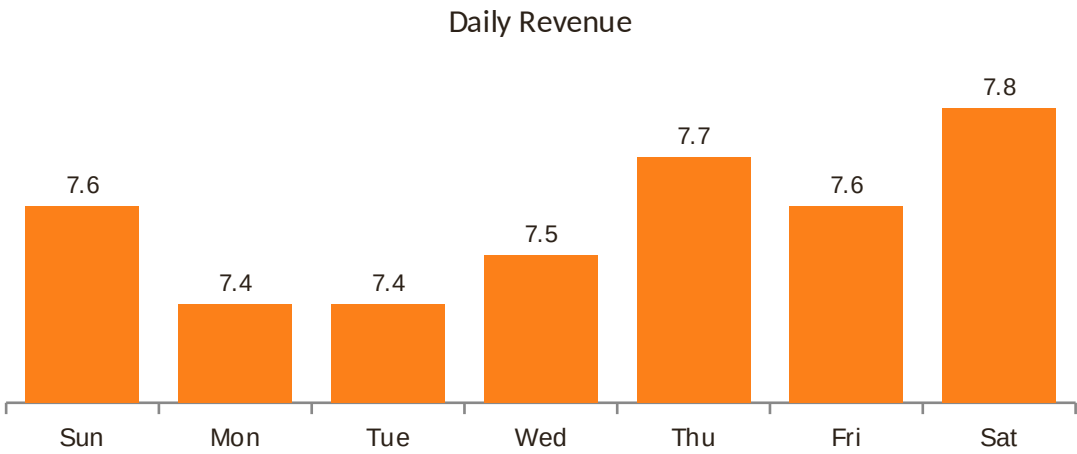
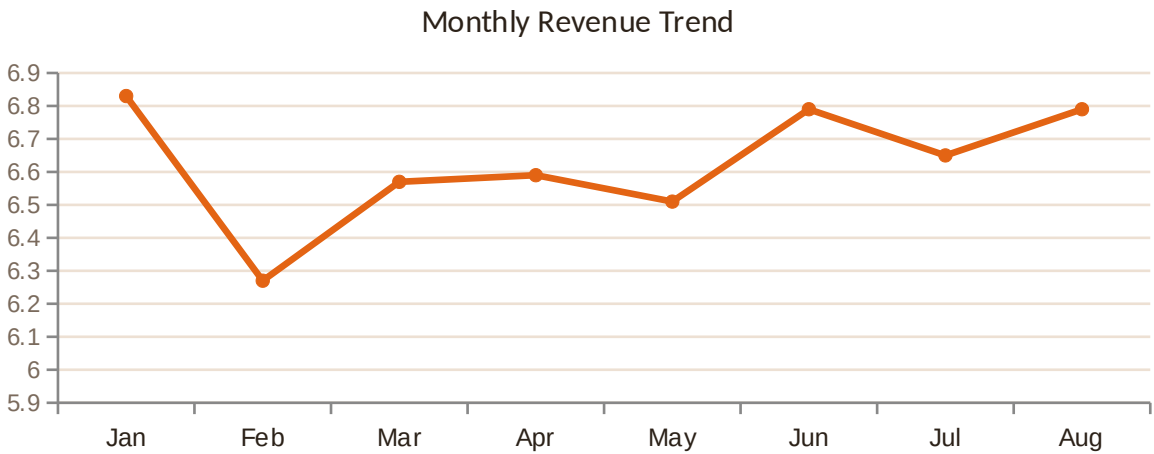
total_sales
57.44 INR Millions

KPI Queries

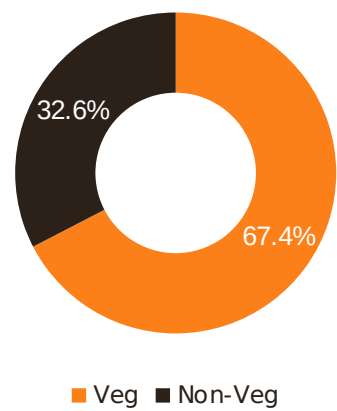
Typical validation checks: orders without matching customers/restaurants, negative order amounts, delivery time earlier than pickup time, and referential integrity across the star schema (fact_orders ↔ dim_restaurant, dim_location, dim_dish, dim_date).

Python Analysis

Exploratory data analysis in pandas — merging, feature engineering, and trend discovery.



Revenue by Food Category

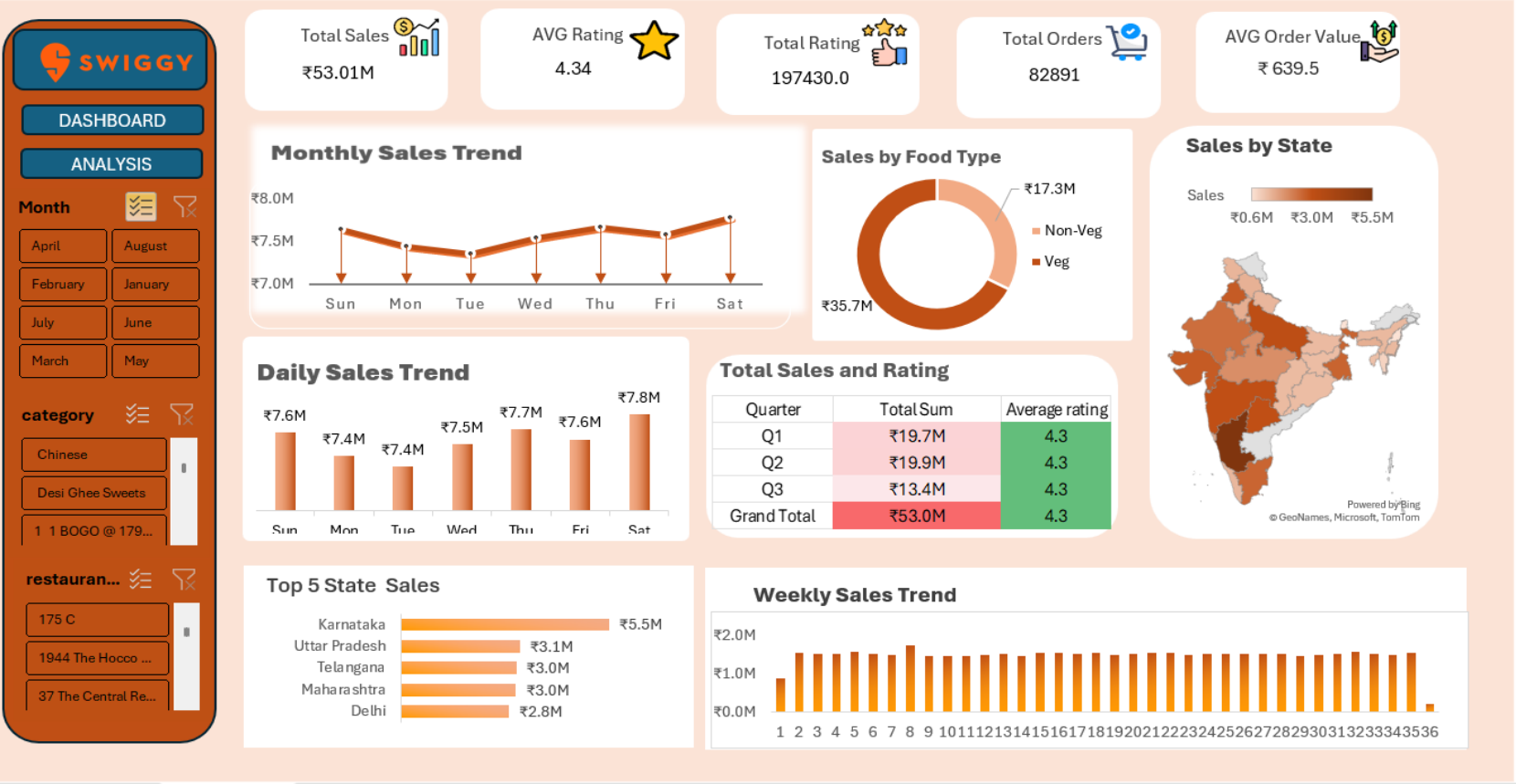


Pipeline Steps (pandas)

- Merged fact_orders with dim_restaurant, dim_location, dim_dish, dim_date
- Engineered Year, Month, Quarter, Day features from order_date
- Classified dishes into Veg / Non-Veg via keyword matching
- Computed KPIs: total sales, AOV, avg rating, order count

Excel Analysis

Pivot-table driven dashboard with quarter and day-level detail.



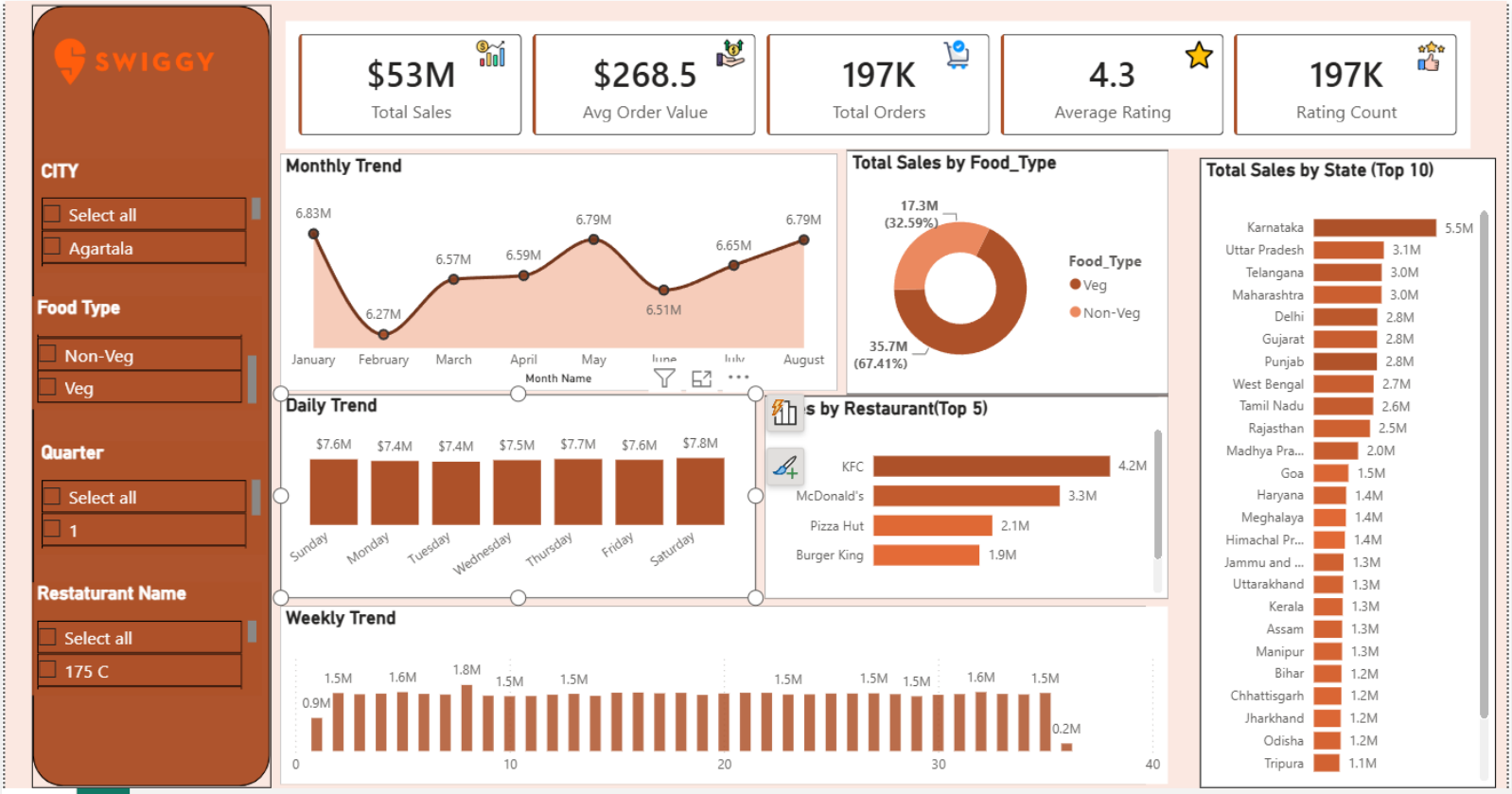
What's In It

- Data model linking orders to restaurant, location, and date dimensions
- Pivot tables & pivot charts for daily, weekly, and monthly sales
- Quarterly KPI table: total sales & average rating per quarter
- Slicers for month, category, and restaurant-level filtering

Key Excel KPIs: Grand Total ₹53M sales, Total Orders 24,588 (Q1 slice), Average Rating 4.34, held steady at 4.3 across every quarter.

Power BI Dashboard

The full interactive report — filterable by city, food type, quarter, and restaurant.



\$53M

Total Sales

197K

Total Orders

\$268.5

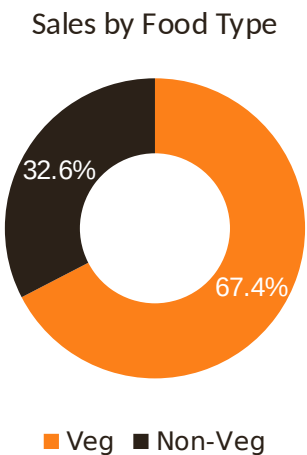
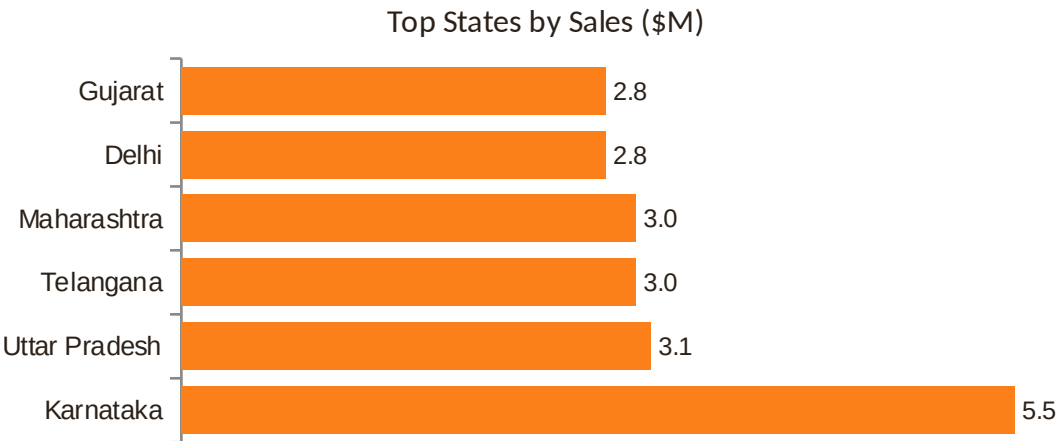
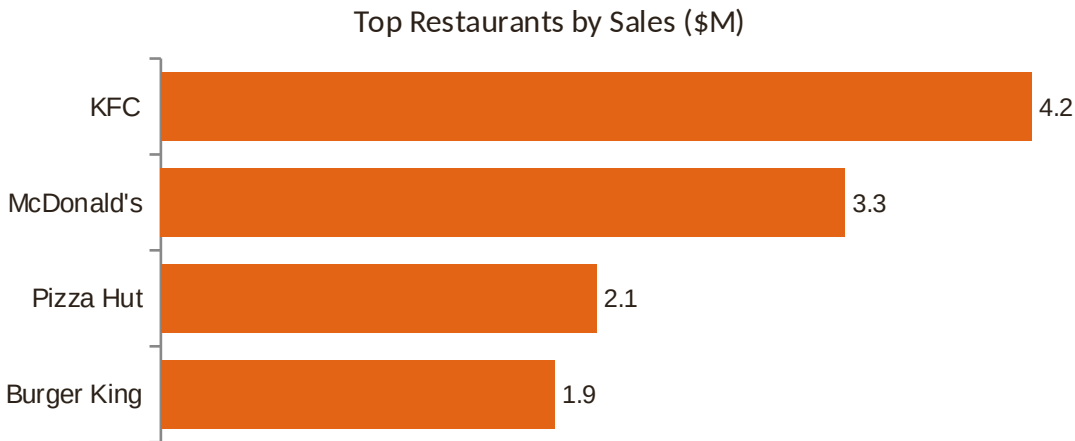
Avg Order Value

4.3 ★

Avg Rating

Power BI Dashboard — Category & Location Insights

Extracted views from the report for restaurant, state, and food-type performance.



Reading the Cuts

- KFC and McDonald's alone out-sell the other top brands combined
- Karnataka nearly doubles the next closest state
- Veg orders outnumber non-veg roughly 2 to 1 across the platform

Key Insights

What the data tells us.



Top restaurants lead the pack

KFC (\$4.2M) and McDonald's (\$3.3M) are the top 2 restaurant brands by sales — together outselling the next four brands combined.



Weekends outperform weekdays

Saturday sales peak at \$7.8M, roughly 5% above the weekly low on Monday/Tuesday (\$7.4M).



Karnataka is the strongest state

At \$5.5M, Karnataka nearly doubles the next closest state, Uttar Pradesh (\$3.1M).



Veg dominates demand

67% of revenue comes from veg dishes, vs. 33% for non-veg — a 2:1 split.



Ratings held steady all year

Average rating stayed at 4.3 in every quarter, even as sales volumes shifted.

Skills Demonstrated

The toolkit behind this project.



SQL



Python



Data Cleaning



EDA



Excel



Power BI



Data Modeling



Dashboarding

Conclusion

What I Learned

- How to build an end-to-end analytics pipeline — from raw multi-table data to a polished dashboard
- The importance of validating one consistent source of truth across Python, Excel, and Power BI
- How small data-quality issues (duplicates, nulls, referential integrity) cascade into dashboard accuracy if not caught early

Business Impact

- Gives stakeholders a self-serve view of sales, restaurant, and location performance
- Surfaces where to focus promotions — weekend timing, top-performing states like Karnataka
- Confirms service quality held steady (4.3★) even as demand fluctuated quarter to quarter

Built end-to-end: SQL → Python → Excel → Power BI

Thanks for reading — let's connect! [Yamini.Sudulaguntla] | www.linkedin.com/in/YaminiSudulaguntla